Neosha Narayanan

Atlanta, GA | neosha@gatech.edu | neosha@alum.mit.edu | neoshanarayanan.github.io

About

I'm a doctoral candidate at Georgia Tech studying the hydrology of glaciers and ice sheets using both geophysical observations and large-scale ice sheet models. My doctoral research addresses both (1) observing the pathways meltwater takes from the surface to the bed, and (2) modeling the effects of water injections at the ice-bed interface on glacier motion. As a co-chair of the US Association for Polar Early Career Scientists (USAPECS), I'm also committed to advocating for young polar researchers who will shape the future of our field.

Education

Georgia Institute of Technology — Expected Spring 2028
Ph.D. in Geophysics — Advised by Dr. Winnie Chu — *Atlanta, GA*

- Scholarships and Awards:
  - Evans Family Fellow
  - Scheller College Graduate Sustainability Fellow

Massachusetts Institute of Technology (MIT) — July 2023
Master of Science in Geophysics (GPA: 4.8/5.0) – Advised by Dr. Brent Minchew — *Cambridge, MA*

- Thesis title: Modeling Subglacial Hydrology in the Himalayas
- Scholarships and Awards:
  - William Asbjornsen Albert Memorial Fellow

Massachusetts Institute of Technology (MIT) — May 2022
Bachelor of Science in Materials Science and Engineering (GPA: 4.8/5.0) — *Cambridge, MA*

- Minors in Writing and South Asian Studies
- Leadership, Scholarships, and Awards:
  - President, South Asian Association of Students (SAAS) (2021-2022)
  - Priscilla King Gray (PKG) Fellow (2021)
  - Burchard Scholar (2020)
  - Henry David Thoreau Scholar (2018-2022)
  - Cycling Club Recruitment Officer (2020-2021)

Amherst Regional High School — June 2018
Valedictorian — *Amherst, MA*

Upcoming Publications

Ishraque, F., Shadab, M.A., Shionalyn, K., **Narayanan, N.**, et al. **(in review)**. Assessing Challenges for U.S.-Based Polar Early Career Scientists During Science Policy Changes. September 13, 2026. *Journal of Glaciology.* https://doi.org/10.31223/X5NR5S

Sommers, A.N., **Narayanan, N.**, Steiner, J.F., Vijay, S., and Meyer, C.R. (**in prep.**) Localized Glacier Acceleration Emerges from Coupled Subglacial Hydrology and Ice Flow Modeling. *Geophysical Research Letters.*

Narayanan, N., Sommers, A.N., Robel, A., Meyer, C.R., Chu, W. **(in prep.)** Ice-dammed Lake Outburst May Assist Glacier Surge Termination in the Karakoram.

Publications

4. **Narayanan, N.**, Sommers, A., Chu, W., Steiner, J., Siddique, A.M., Meyer, C., Minchew, B. (2025). Simulating Seasonal Evolution of Subglacial Hydrology at a Surging Glacier in the Karakoram. *Journal of Glaciology.* doi.org/10.1017/jog.2025.10078

3. **Narayanan, N.** (2023). Modeling Subglacial Hydrology in the Himalayas. Master's thesis, Massachusetts Institute of Technology. https://hdl.handle.net/1721.1/153347

2. Su, I., **Narayanan, N.**, et al. In-situ spider web construction and mechanics (2021). *Proceedings of the National Academy of Sciences (PNAS).*

1. Su, I., Jung, G.S., **Narayanan, N.**, Buehler, M. Perspectives on 3D printing of self-assembling materials and structures (2020). *Current Opinion in Biomedical Engineering.*

Conference Proceedings

14. **Narayanan, N.**, Christoffersen, M., Dawson, E., Chu, W., Valentine, A., Chivers, C., Poinar, K., Meyer, C.R. (2026, December) Spatial heterogeneity in Helheim's firn aquifer revealed by ground-penetrating radar. American Geophysical Union Fall Meeting. (POSTER).

13. Wright, B., **Narayanan, N.**, Hoffman, A., Chu, W. (2026, December) Evaluating the Effect of a Soft Basal Ice Unit on Glacier Flow. American Geophysical Union Fall Meeting (POSTER).

12. Ishraque, F., Shadab, M.A., Shionalyn, K., **Narayanan, N.**, et al. (2026, December). Assessing Challenges for U.S.-Based Polar Early Career Scientists During Science Policy Changes. American Geophysical Union Fall Meeting (POSTER).

11. **Narayanan, N.**, Christoffersen, M., Dawson, E., Chu, W., et al. (2026, September). New radar-derived insights into the firn aquifer at Helheim Glacier. International Glaciological Society (IGS) Radioglaciology Symposium, Tübingen, Germany (ORAL PRESENTATION).

10. **Narayanan, N.**, Chu, W., Sommers, A., Meyer, C., Steiner, J., Minchew, B. (2026, April). Investigating Relationships between Lake Drainages and Surge Motion through Modeling in the Karakoram Himalaya. European Geosciences Union General Assembly (POSTER).

9. **Narayanan, N.**, Chu, W., Sommers, A., Meyer, C., Steiner, J., Robel, A., Minchew, B. (2025, December). Coupled Subglacial Hydrology and Ice Dynamics Modeling of Lake Drainages and Surging Glaciers in the Himalayas. American Geophysical Union Fall Meeting (ORAL PRESENTATION).

8. Ishraque, F., Afzal Shadab, M., Lauter, O., **Narayanan, N.**, et al. (2025, December). Assessing Challenges for Polar Early Career Scientists During Science Policy Upheaval (invited). American Geophysical Union Fall Meeting.

7. **Narayanan, N.**, Chu, W., Meyer, C., Poinar, K., Sommers, A., Mejia, J., et al. (2024, December). Radar Measurements of Firn Aquifer at Helheim Glacier, Greenland. American Geophysical Union Fall Meeting (POSTER).

6. Poinar, K., Sommers, A., Mejia, J., Meyer, C., **Narayanan, N.**, et al. (2024, December). Hydrologic Implications of the Water Feeding Ice Caves at Helheim Glacier, Greenland. American Geophysical Union Fall Meeting (POSTER).

5. Mejia, J., Poinar, K., Meyer, C., Sommers, A., **Narayanan, N.**, Chu, W. (2024, December). Direct Observations of Mixed-Mode Fracture Opening of a Water-Fed Crevasse on Helheim Glacier, Greenland. Amerian Geophysical Union Fall Meeting (POSTER).

4. **Narayanan, N.**, Sommers, A., Chu, W., Steiner, J., Siddique, A.M., Meyer, C., Minchew, B. (2024, April). Coupled Ice Dynamics and Subglacial Hydrology Simulations of Glacier Surges in the Himalayas. European Geosciences Union (POSTER). https://doi.org/10.5194/egusphere-egu24-11819

3. **Narayanan, N.**, Sommers, A., Steiner, J., Siddique, A.M., Minchew, B. (2023, December). Modeling Subglacial Hydrology in the Himalayas: Applications to Surges and Glacial Lake Outburst Floods. American Geophysical Union Fall Meeting. (POSTER)

2. **Narayanan, N.**, Millstein, J., and Minchew, B. (2022, December). Simulation and Analysis of Deformation and Stability in Antarctic Ice Shelves. American Geophysical Union Fall Meeting. (POSTER)

1. **Narayanan, N.**, Millstein, J., and Minchew, B. (2022, April). Simulation and Analysis of Deformation in Antarctic Ice Shelves. Northeast Glaciology Meeting. (POSTER)

Invited Talks

Coupled Subglacial Hydrology Modeling of Lake Drainages | Maths on Ice — December 11, 2025
- Coupled Subglacial Hydrology and Ice Dynamics Modeling of Lake Drainages and Glacier Motion in the Himalayas

The Cool Science of Glaciers! | Spectacular Science Podcast with Akshay Jayaraman — October 2024

Geophysics Department Seminar | Georgia Tech School of Earth and Atmospheric Sciences — April 2024

International Women's Day Panelist | MIT South Asian Alumni Association — April 2023

Keynote Speaker | Henry David Thoreau Society Annual Meeting — April 2022

"Intuitively Obvious" AAPI student moderator | MIT — 2022

Outreach and Professional Activities

Manuscript referee, npj Natural Hazards and Scientific Advances — 2026-Present
Manuscript referee, The Cryosphere — 2025-Present
Co-chair, US Association for Polar Early Career Scientists (USAPECS) — 2025-Present
Vice President, Georgia Tech Graduates in Earth and Atmospheric Sciences (GEAS) — 2025-2026
Georgia Tech Clough Climate Seminar Committee — 2025-Present
Georgia Tech EAS Graduate Research Symposium Chair — 2025
USAPECS Board Member — 2024-2025
Letters to a Pre-Scientist STEM Professional Volunteer — 2023-2025
MIT Admissions Alumni Educational Counselor (EC) — 2022-Present

Grants and Proposals

NSF OPP (submitted, student collaborator): Collaborative Research: Quantifying, Visualizing, and Modeling the Formation, Evolution, and Impact of Greenland Basal Ice (2026)

Arête Glacier Initiative (submitted as early career Principal Investigator): Evaluating radar-derived controls on basal rheology and internal ice deformation. (2026) - not funded

NASA ROSES: Understanding changes in High Mountain Asia, "Quantifying the role of subglacial hydrology in glacier surges and outburst floods of High Mountain Asia" (2024-present). Contributed significantly to grant ideation and writing as a MS student. Awarded to Dartmouth College.

Fieldwork

Follow the Water: Hydrology at Helheim Glacier | Tasiilaq, Greenland (Georgia Tech) — June 2026
- Part of a collaborative expedition on the Greenland Ice Sheet.
- Carried out common offset and common midpoint surveys using PulseEkko ground-penetrating radar towed on snowmobiles.
- Led sling-loading ground operations for helicopter transport of snowmobiles.

Greenland Ice-Sheet and Ocean Summer School | Nuuk, Greenland — May 2026
- Selected as 1 of 12 participants in a highly competitive, 2-week summer school for mid- and late-stage PhD students.
- Engaged in discussions about ice-ocean interactions, creative scientific thinking, and collaborative science on Indigenous land.
- Produced a 17-minute podcast connecting GRISO participant science with societal relevance. Worked with a group of two other participants.

Follow the Water: Hydrology at Helheim Glacier | Tasiilaq, Greenland (Georgia Tech) — July 2025
- Part of a collaborative expedition on the Greenland Ice Sheet. Deployed radar echo-sounders (ApRES), ground-penetrating radar (PulseEKKO and Groundhog)
- Carried out common offset, common midpoint, and multi-offset surveys
- Successfully carried out retrieval of subsurface UNAVCO GPS units

Follow the Water: Hydrology at Helheim Glacier | Tasiilaq, Greenland (Georgia Tech) — July 2024
- Deployed ApRES and seismo-electric instruments, dug snowpits, and assisted with equipment retrieval

- Analyzed near-surface ApRES data using xApRES and impDAR, carried out least-squares and MCMC inversion for near-surface firn structures.

International Summer School in Glaciology | McCarthy, Alaska June 2024
- Summer school participant. Learned the foundations of glaciology through 2 weeks of hands on lectures.
- Wrote a numerical model for distributed subglacial hydrology in Python and Julia.

Geology and Sedimentology Field Course | MIT-EAPS Crosby Trip, Sicily, Italy June 2023
- Learned about geology and sedimentology through hands-on activities.

Glacial Lake Survey Study with IIT Roorkee | Himachal Pradesh, India July 2022
- Conducted a qualitative field survey with scientists from the Indian Institute of Technology, Roorkee.

MIT CEE Traveling Research Experience (TREX) | Hawaii, US January 2020
- Conducted forest transects and volatile analysis. Analyzed invasive plant species and air quality.

Geology First-Year Orientation Program Teaching Assistant | Yellowstone National Park August 2019
- Taught incoming first-year undergraduates about geology through a hands-on environment.
- Assisted with logistics and transportation of first-year undergraduates.

Teaching and Mentoring

Undergraduate Research Mentoring Present
Georgia Institute of Technology *Atlanta, GA*
- Brooklyn Wright (REU, Columbia University, Summer 2026): Investigating the Effect of Soft Basal Ice on Ice Flow
  - Accepted as an abstract to AGU fall meeting in 2026.

MIT Educational Counselor 2022-Present
Massachusetts Institute of Technology *Amherst, MA*
- Interviewing and mentoring MIT undergraduate applicants from western Massachusetts.

Graduate Teaching Assistant 2023
Georgia Institute of Technology *Atlanta, GA*
- EAS 2600: Earth Processes
  - Lab TA in Fall 2023
  - Wrote and delivered lectures on introductory earth science topics, guided students through hands-on earth science lab assignments, held office hours to assist with conceptual questions, and graded lab assignments.

Terrascope Undergraduate Teaching Fellow (UTF) 2020-2022
Massachusetts Institute of Technology (MIT) *Cambridge, MA*
- Mentored students, held office hours, and ran meetings for fall *Solving Complex Problems* class which focuses on a different sustainability or climate-related issue each year.
- Evaluated and provided feedback on student-produced website and presentation drafts prior to final presentations.
- Collaborated with other UTFs to research and develop innovative methods for remote collaborative project work during COVID-19 pandemic.

Associate Advisor 2019-2022
Massachusetts Institute of Technology (MIT) *Cambridge, MA*
- Assisted faculty with advising first-year undergraduates by providing advice on class registration and helping students find resources for academic performance and mental health.
- Mentored 21 total first-year students both in-person and during the COVID-19 pandemic through weekly one-on-one meetings.

Research and Work Experience

Indian Institute of Technology, Roorkee 2022

Visiting Researcher: Department of Hydrology *Roorkee, Uttarakhand, India*

- Advisor: Dhyan Singh Arya
- Performed literature review of glacial lake outburst flooding (GLOF) mechanisms in the Himalayas.
- Identified vulnerable flood-prone areas using literature review and remote sensing data.
- Developed an inventory of glacier-proximal lakes in the Koshi River Valley using QGIS, evaluated existing risk analysis protocols, and performed HEC-RAS modeling of glacial lake outburst flood (GLOF) flow paths.
- Conducted a field study of Suraj Tal lake in Himachal Pradesh.

Glacier Dynamics and Remote Sensing Group 2020-2021

Undergraduate Researcher: MIT Department of Earth, Atmospheric, and Planetary Sciences *Cambridge, MA*

- Advised by Joanna Millstein and Brent Minchew.
- Wrote a numerical solver for 2-dimensional Stokes flow to analyze the stress field induced by a pinning point at an idealized Antarctic ice shelf.
- Developed a new metric for predicting ice shelf stability based on stress and velocity of ice shelves.

Charles River Watershed Association 2021

Data Analysis Intern *Cambridge, MA*

- Compiled and evaluated stormwater reporting and water quality data from watershed communities.
- Translated results of survey into a public-facing report to encourage municipal compliance with the Massachusetts Small Municipal Separate Storm Sewer System (MS4) Permit.
- Developed recommendations for long-term data tracking and analysis of phosphorus pollution in the Charles River watershed.

MIT Angles Magazine 2021

Editorial Assistant *Cambridge, MA*

- Selected by MIT Writing department faculty as a student editor for MIT online magazine highlighting student writing.
- Coordinated communication between faculty editors and student writers, edited writer biographies, proofread and edited student writing pieces, and formatted magazine website.

MIT-South Asia Oral History Project 2020-2021

Student Intern, MIT History Department *Cambridge, MA*

- Studied literature on the history of Indian students at MIT and its relationship with technical institutions in India during British occupation and India's independence movement.
- Obtained and read through MIT student newspaper archives and graduation records and wrote articles about early South Asian MIT students as part of a larger exhibit on South Asia and MIT.

Laboratory for Atomistic and Molecular Materials 2019-2020

Undergraduate Researcher: MIT Department of Civil and Environmental Engineering *Cambridge, MA*

- Advised by Isabelle Su and Markus Buehler.
- Designed and built an Arduino-controlled laser scanning system to develop digital models of 3-dimensional spiderwebs.
- Developed methods for breaking down post-consumer PLA plastic and and extruding it for high-quality fused deposition modeling (3-D printing) of recycled plastic and biomass waste with enhanced mechanical properties.
- Carried out tensile strength and conductivity testing of 3-D printed composite materials.

Susan Solomon Lab 2018

Undergraduate Researcher: MIT Department of Earth, Atmospheric, and Planetary Sciences *Cambridge, MA*

- Obtained and analyzed satellite data on atmospheric ozone, pollutants, and aerosols using MATLAB.